

University
Politehnica
of Bucharest





Computer
Science and
Engineering
Department

Document-Oriented Databases
MongoDB (I)

Ș.L. Dr. Ing. Ciprian-Octavian Truică
ciprian.truica@cs.pub.ro



Overview

- Document-Oriented Databases
- MongoDB



Overview

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- MongoDB

Document-Oriented Databases

- Document-Oriented Databases Management Systems (DODBMS) have evolved from Key-Value Databases
- DODBMS are used for storing, retrieving and managing semi-structured data
- DODBMS propose a schema-less flexible data representation for data providing more flexibility for data modeling



Document-Oriented Databases

- DODBMS use documents for storing data
- The term document may usually refer a Word or PDF documents, but in this case refers to JSON or XML files
- The flexibility provided by XML and JSON make it easier to modify documents than it is for tables in a RDBMS
- Documents are stored in collections

Document-Oriented Databases

- In DODBMS information is stored using labels to describe the data and values for that description

- New information (labels) can be added directly to a record without the need to modify the entire schema as it is for RDBMS



RDBMS vs DODBMS Comparison

RDBMS	DODBMS
Database (Schema)	Database
Table	Collection
Attribute	Field (tag, description)
Attribute values	Field (tag, description) value

RDBMS vs DODBMS Comparison





```
{     customer_id : 1,
      first_name : "Mark",
      last_name : "Smith",
      city : "San Francisco",
      phones: [ {
            type : "work",
            number: "1-800-555-1212"
      },
      {     type : "home",
            number: "1-800-555-1313",
            DNC: true
      },
      {     type : "home",
            number: "1-800-555-1414",
            DNC: true
      }
   ]
}
```



Benefits of a DODBMS

- Flexible data modeling
- Fast write performance
- Fast query performance



Benefits of a DODBMS
Flexible data modeling

- Data from the web, mobile, social and IoT applications change the nature of the application's data model

- In a RDBMS, this changes impose the modification of the schema by altering tables and adding or removing new columns

- The Flexibility of DODBMS eliminate the need to force-fit the data into predefined attributes and tables



Benefits of a DODBMS
Fast write performance

- Some DODBMS prioritize write availability over strict data consistency
- This ensures that writes will always be executed even if there is a hardware or network failure
- In case of failure, the replication and eventual consistency mechanisms ensures that the environment will function



Benefits of a DODBMS
Fast query performance

- Most DODBMS
  - Provide powerful query engines for CRUD operations
  - Use indexes and secondary indexes to improve data retrieval
  - Provide aggregation frameworks for Data Analysis and Business Intelligence
    - Aggregation is either native or done through MapReduce



Overview

- Document-Oriented Databases
- MongoDB
  - Overview
  - Storage engine: MMAPv1
  - Storage engine: WiredTiger



MongoDB – Overview

- MongoDB (from humongous) is focused on combining the critical capabilities of RDBMS with the innovations of NoSQL DBMS
- MongoDB is an open-source Document-Oriented Database Management System (DODBMS)
- MongoDB is written in C++, C and JavaScript
- Run on different OS: Linux, Windows, Solaris, Mac OSX, FreeBSD
- Can be easily deployed in the Cloud as a DaaS



MongoDB – Overview

- MongoDB uses a flexible, dynamic schema to store data
- A MongoDB instance is started using mongod
- A record is stored in a document
- Multiple documents are stored in a collection



MongoDB – Overview

- Documents in a collection do not necessarily have the same structure:
  - The number of attributes can differ from one record to another
  - The fields' data type can differ from one record to another

- Although the database allows documents with a different number of attributes and for the same fields different data types, in the same collection records have almost the same structure

- In practice, documents usually model objects from a high-level programming language



MongoDB
Data representation

- MongoDB stores the data in BSON documents
- BSON is a binary-encoded serialization of JSON-like documents
  - Note: Serialization is the process of translating data structures or object state into a format that can be stored (e.g. in a file, memory buffer, etc.) or transmitted (e.g. across a network connection link, etc.) and reconstructed later (possibly in a different computer environment)
- A BSON document is limited in size to 16MB
- Solution to the size limitation MongoDB GridFS



MongoDB
Data representation

- The main features of BSON are:
  - Lightweight – for fast retrieval of information over a computer network it is necessary to keep the overhead needed to store data to a minimum
  - Traversable – it is import to store data in a format that is easily parsed
  - Efficient – using C language data types, data is easily encoded and decoded



MongoDB
Key Features

- Support of multiple storage engines
- High availability through replication
- Automatic horizontal scaling though sharding
- High performance
- Rich query language



MongoDB
Key Features

- MongoDB supports multiple storage engines:
  - MMAPv1 Storage Engine (deprecated starting MongoDB v4)
  - WiredTiger Storage Engine



MongoDB
Key Features

- High performance
  - Support for embedded data models reduces I/O activity on database systems:
    - Supports nested documents
    - References can also be used
  - Indexes support faster queries
  - Indexes can include keys from embedded documents and arrays.
  - Indexes can be added on text and geospatial data
  - MongoDB support compound indexes



MongoDB
Key Features

- Note on Indexes:
  - Index prefix is the label that defines the index, e.g.:

  {idx_1: 1}

  - Compound indexes use multiple fields in their structure, e.g.:

  {idx_1: 1, idx_2: 1, idx_3: 1}

  - Queries that use efficiently a compound index use the labels in order, e.g.
    - idx_1,
    - idx_1 and idx_2
    - idx_1 and idx_2 and idx_3



MongoDB
Key Features

- Note on Indexes:
  - Also works for our example:
    - idx_1 and idx_3
  - Not as efficient as creating a compound index only on idx_1 and idx_3
  - Queries that don't use the first label(s) (i.e. idx_1), don't use the compound index
    - idx_2
    - idx_3
    - idx_2 and idx_3
    - idx_3 and idx_2



MongoDB
Key Features

- Rich query language
  - Supports all the CRUD Operations
  - Supports data aggregation through MapReduce and Aggregation Pipeline
  - Supports Text Search and regular expressions
  - Supports Geospatial queries



MongoDB
Key Features

- High availability
  - Supports replication, called Replica Set, which provide:
    - Automatic failover
    - Data redundancy
    - Increased data availability
  - A Replica Set is a group of MongoDB servers that maintain the same data set



MongoDB
Key Features

- Horizontal Scalability
  - MongoDB provides horizontal scalability as part of its core functionality
  - Scalability is achieved through Sharding
  - Sharding distributes data across a cluster of machines
  - Form MongoDB 3.4 the support of creating zones of data on the shard key was added.
  - Using zones in a balanced cluster, MongoDB directs reads and writes covered by a zone only to those shards inside the zone



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MongoDB – MMAPv1

- MMAPv1 Storage Engine
  - The original storage engine
  - It excels at workloads with high volume of inserts, reads and in-place updates
  - Features:
    - Journal
    - Record storage characteristics
    - Record allocation strategies
    - Memory use



MongoDB – MMAPv1 Journal

- To ensure data persistence, all modifications are registered to an on-disk Journal
- MongoDB writes more frequently to the Journal then it writes to the data files
- In the default configuration of the MMAPv1 there are Journal writes roughly every 100ms and data file writes every 60s
- To change the write intervals change the values for the following parameters:
  - Journal file : storage.journal.commitIntervalMs
  - Data file: storage.syncPeriodSecs



MongoDB – MMAPv1
Journal

- The interval values represent the maximum amount of time between
  - The completion of a write operation and
  - When MongoDB writes to the data files or to the journal files

- In many cases MongoDB and the operating system flush data to disk more frequently (the above values represents a theoretical maximum)

- The journal allows MongoDB to successfully recover data from data files after a mongod instance exits without flushing all changes



MongoDB – MMAPv1
Record allocation strategies

- MongoDB supports multiple record allocation strategies that determine how mongod adds padding to a document when creating a record

- Because documents in MongoDB may grow after insertion and all records are contiguous on disk, the padding can reduce the need to relocate documents on disk following updates

- Relocations
  - Are less efficient than in-place updates
  - Can lead to storage fragmentation

- As a result, all padding strategies trade additional space for
  - Increased efficiency
  - Decreased fragmentation



MongoDB – MMAPv1
Record allocation strategies

- There are two allocations strategies
  - Power of 2 allocations
  - No padding allocation strategy (or exact fit allocations)

- The two different allocation strategies support different kinds of workloads:
  - The power of 2 allocations are more efficient for insert/update/delete workloads
  - Exact fit allocations is ideal for collections without update and delete workloads



MongoDB – MMAPv1
Record storage characteristics

- MongoDB uses Power of 2 Sized Allocations by default
- Every document in MongoDB is stored in a record which contains:
  - The document itself
  - Extra space, or padding
- Padding
  - Allows the document to grow as the result of updates
  - Minimizing the likelihood of reallocations



MongoDB – MMAPv1
Power of 2 Sized Allocations

- Appeared with MongoDB 3.0
- Uses the power of 2 sizes allocation as the default record allocation strategy for MMAPv1
- With this strategy, each record has a size in bytes that is a power of 2 (e.g. 32, 64, 128, 256, 512 … 2 MB)
- For documents larger than 2 MB, the allocation is rounded up to the nearest multiple of 2 MB



MongoDB – MMAPv1
Power of 2 Sized Allocations

- Properties
  - Can efficiently reuse freed records to reduce fragmentation:
    - Quantizing record allocation sizes into a fixed set of sizes increases the probability that an insert will fit into the free space created by an earlier document deletion or relocation.
  - Can reduce moves:
    - The added padding space gives a document room to grow without requiring a move.
    - In addition to saving the cost of moving, this results in less updates to indexes.
    - Although the power of 2 sizes strategy can minimize moves, it does not eliminate them entirely.



MongoDB – MMAPv1

No Padding Allocation Strategy

- Is used for collections whose workloads do not change the document sizes
- Workloads consist of
  - Insert-only operations
  - Update operations that do not increase document size
- Prior MongoDB v3.0.0, the allocation strategy included a dynamically calculated padding as a factor of the document size



MongoDB – MMAPv1
Memory use

- MongoDB automatically uses all free memory on the machine as its cache
- System resource monitors show that MongoDB uses a lot of memory (its usage is dynamic)
- If another process suddenly needs half the server's RAM, MongoDB will yield cached memory to the other process



MongoDB – MMAPv1
Memory use

- The operating system's virtual memory subsystem manages MongoDB's memory
- This means that MongoDB will use as much free memory as it can, swapping to disk as needed
- Deployments with enough memory to fit the application's working data set in RAM will achieve the best performance



Overview

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MongoDB – WiredTiger

- The default storage for the latest versions
- Introduced from MongoDB v3.0
- Features:
  - Document level concurrency
  - Snapshots and checkpoints
  - Journal
  - Compression
  - Memory use



MongoDB – WiredTiger
Document level concurrency

- WiredTiger uses document-level concurrency control for write operations

- Multiple clients can modify different documents of a collection at the same time



MongoDB – WiredTiger
Document level concurrency

- WiredTiger uses optimistic concurrency control for most read and write operations.

- WiredTiger uses only intent locks at the:
  - Global level
  - Database level
  - Collection levels

- When the storage engine detects conflicts between two operations, one will incur a write conflict causing MongoDB to transparently retry that operation



MongoDB – WiredTiger
Document level concurrency

- Global "instance-wide" lock is required in general for some global operations
  - E.g. short-lived operations involving multiple databases
- Exclusive database lock is required for other operations,
  - E.g. dropping a collection



MongoDB – WiredTiger
Snapshots and checkpoints

- WiredTiger uses MultiVersion Concurrency Control (MVCC)
  - Also available for other databases, e.g. Couchbase

- WiredTiger provides a point-in-time snapshot of the data to the transaction at the start of an operation

- A snapshot presents a consistent view of the in-memory data



MongoDB – WiredTiger
Snapshots and checkpoints

- When writing to disk, WiredTiger writes all the data in a snapshot to disk in a consistent way across all data files
- The now-durable data act as a checkpoint in the data files
- The checkpoint ensures that the data files are consistent up to and including the last checkpoint (i.e. checkpoints can act as recovery points)
- MongoDB configures WiredTiger to create checkpoints (i.e. write the snapshot data to disk) at intervals of 60 seconds or 2 gigabytes of journal data



MongoDB – WiredTiger
Snapshots and checkpoints

- During the write of a new checkpoint, the previous checkpoint is still valid
- As such, even if MongoDB terminates or encounters an error while writing a new checkpoint, upon restart, MongoDB can recover from the last valid checkpoint
- The new checkpoint becomes accessible and permanent when WiredTiger's metadata table is atomically updated to reference the new checkpoint
- Once the new checkpoint is accessible, WiredTiger frees pages from the old checkpoints



MongoDB – WiredTiger
Snapshots and checkpoints

- MongoDB can recover from the last checkpoint when using WiredTiger, even without journaling

- However, to recover changes made after the last checkpoint, MongoDB should be run with journaling



MongoDB – WiredTiger Journal

- WiredTiger uses checkpoints to
  - Provide a consistent view of data on disk
  - Allow MongoDB to recover from the last checkpoint

- However, if MongoDB exits unexpectedly in between checkpoints, journaling is required to recover information that occurred after the last checkpoint



MongoDB – WiredTiger Journal

- The recovery process with journaling:
  - Looks in the data files to find the identifier of the last checkpoint
  - Searches in the journal files for the record that matches the identifier of the last checkpoint
  - Apply the operations in the journal files since the last checkpoint



MongoDB – WiredTiger Journaling Process

- WiredTiger creates one journal record for each client initiated write operation
- The journal record includes any internal write operations caused by the initial write
- WiredTiger creates a single journal record that includes both the update operation and its associated index modifications



MongoDB – WiredTiger Journaling Process

- MongoDB configures WiredTiger to use in-memory buffering for storing the journal records
- Threads coordinate to allocate and copy into their portion of the buffer
- All journal records up to 128 kB are buffered



MongoDB – WiredTiger Journaling Process

- WiredTiger syncs the buffered journal records to disk according to the following intervals or conditions:
  - Every 50 milliseconds from MongoDB v3.2
  - MongoDB sets checkpoints to occur in WiredTiger on user data at an interval of 60 seconds or when 2 GB of journal data has been written, whichever occurs first.
  - WiredTiger forces a sync of the WiredTiger journal files if the write operation includes a write concern that requests the acknowledgement from MongoDB that the write operation has been written to the journal



MongoDB – WiredTiger Journaling Process

- WiredTiger syncs the buffered journal records to disk according to the following intervals or conditions:
  - When WiredTiger creates a new journal file, WiredTiger syncs the previous journal file
  - WiredTiger creates a new journal file approximately every 100 MB of data
    - This happens because MongoDB uses a journal file size limit of 100 MB



MongoDB – WiredTiger Journal Files

- For the journal files, MongoDB creates a subdirectory named journal under the database directory (in this directory the data is stored)
- WiredTiger journal files have names with the following format WiredTigerLog.<sequence> where <sequence> is a zero-padded number starting from 0000000001
- Journal files contain a record per each write operation
- Each record has a unique identifier
- MongoDB configures WiredTiger to use snappy compression for the journaling data
  - Snappy is a fast data compression and decompression library written in C++ by Google based on ideas from LZ77 and open-sourced in 2011.



MongoDB – WiredTiger Journal Files

- Minimum log record size for WiredTiger is 128 bytes
- If a log record is 128 bytes or smaller, WiredTiger does not compress that record
- WiredTiger journal files have a maximum size limit of approximately 100 MB
- WiredTiger creates a new journal file once the file exceeds that limit
- WiredTiger automatically removes old journal files to maintain only the files needed to recover from last checkpoint
- WiredTiger will pre-allocate journal files.



MongoDB – WiredTiger Compression

- WiredTiger supports compression for all collections and indexes
- Compression minimizes storage use at the expense of additional CPU
- By default, WiredTiger uses
  - Block compression with the snappy compression library for all collections
  - Prefix compression for all indexes
- For collections, block compression with zlib is also available



MongoDB – WiredTiger Compression

- To specify an alternate compression algorithm or no compression, use the following setting:

storage.wiredTiger.collectionConfig.blockCompressor

- For indexes, to disable prefix compression, use the following setting:

storage.wiredTiger.indexConfig.prefixCompression

- Compression settings are also configurable on a per-collection and per-index basis during collection and index creation



MongoDB – WiredTiger Compression

- The default compression settings balance storage efficiency and processing requirements for most workloads

- As already mentioned, the WiredTiger journal is also compressed by default



MongoDB – WiredTiger
Memory use

- MongoDB utilizes both the WiredTiger internal cache and the filesystem cache

- Starting with MongoDB v3.4, the WiredTiger internal cache, by default, will use the larger of either:
  - 50% of RAM minus 1 GB, or
  - 256 MB



MongoDB – WiredTiger
Memory use

- Different representations are used for data in the WiredTiger internal cache versus the on-disk format
  1. Data in the filesystem cache is the same as the on-disk format
     - This method benefits of any compression for data files
     - The filesystem cache is used by the operating system to reduce disk I/O
  2. Indexes loaded in the WiredTiger internal cache have a different data representation to the on-disk format
     - This method can still take advantage of index prefix compression to reduce RAM usage
     - Index prefix compression de-duplicates common prefixes from indexed fields



MongoDB – WiredTiger
Memory use

3. Collection data in the WiredTiger internal cache is uncompressed and uses a different representation from the on-disk format
   - Block compression can provide significant on-disk storage savings
   - The data must be uncompressed to be manipulated by the server



Summary

- This course presented:
  - Document-oriented Databases
  - An overview of MongoDB
  - The two storage engines available in MongoDB